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# Applied Cyber Threat Analysis – Email Header Analysis Lab

- This lab was conducted within the Virginia Cyber Range Environment.
  - Report based off assignment assigned by George Mason University - Department of Information Sciences and Technology College of Engineering and Computing
  - Cheche Agada. (2024). Email Header Analysis Lab IT-462: Applied Cyber Threat Analysis, George Mason University
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## Background

The purpose behind this analysis lab is to collect and identify email header information in order to infer if the email in question is malicious in nature.

## Exercise 1

1. What is the Message-ID of the email?

```
DKIM-Signature: v=1; a=rsa-sha256; d=ccrlp.fr; s=mail; c=siml; q=dns;
t=1675600220; h=from:subject:to:date:message-id;
bh=j9mK1OhAKWARs8J8JJkBU5oBEJkOkHngE11dx8Hp+nQ=;
```

The message-id of the email is identified as:

**bh=j9mK1OhAKWARs8J8JJkBU5oBEJkOkHngE11dx8Hp+nQ=**

2. What is the Return-Path of the email?

```
2023-12-05 12:30:28 -0500 Message-ID: <001e61f3ca4f4893984b0fb809690fa4ccrlpfr_117>
Reply-To: COMMUNICATION - CCRLP <communication@ccrlp.fr> Accept-Language: fr-
FR, en-US Content-Language: en-US X-MS-Has-Attach: yes X-MS-TNEF-Correlator:
x-originating-ip: [191.96.227.66] Content-Type: multipart/mixed;
boundary="_006_051e61f3ca4f4893984b0fb809690fa4ccrlpfr_" MIME-Version: 1.0
Return-Path: communication@ccrlp.fr X-MS-Exchange-Organization-
ExpirationStartTime: 05 Feb 2023 12:30:28.9950 (UTC) X-MS-Exchange-
Organization-ExpirationStartTimeReason: OriginalSubmit X-MS-Exchange-
Organization-ExpirationInterval: 1:00:00:00.0000000 X-MS-Exchange-
Organization-ExpirationIntervalReason: OriginalSubmit X-MS-Exchange-
```

The Return-Path of this email is identified as: **communication@ccrlp.fr**

3. What email address was this email sent to?

```
-----
Server id 15.20.6064.34 via Frontend Transport; Sun, 5 Feb 2023 12:30:28
+0000 Received: from talaria2.gmu.edu (talaria2.gmu.edu [129.174.0.123]) by
caduceus2.gmu.edu (Postfix) with ESMTP id 4P8pdJ2Gvlz12b3; Sun, 5 Feb 2023
07:30:28 -0500 (EST) Received: from ironportvm1.gmu.edu (ironportvm1.gmu.edu
-----
```

The email address was sent to: **talaria2.gmu.edu or caduceus2.gmu.edu**

4. What is the X-Sender IP address?

```
313c330u005120; Sun, 5 Feb 2023 07:28:33 -0500 IRONPORT-SDK:
63dfa105_KXX1Joh9FL/OKPmyZLmf7VShoXT0AuDs3DpoIrwV80iqIE
VnfT4TtsyH73o5lexguW6ynL8qTwdyQ9whiJJSw== X-SENDER-IP: 109.69.192.223 X-
```

**X-SENDER-IP: 109.69.192.223**

5. What is the X-Sender-Reputation of the email?

```
SENDER-REPUTATION: 3.5 X-IPAS-Result: =?us-
ascii?q?A0EGCQBun99jh9/ARW1aHAEBaQEBAQcBARIBaQQEAQGBf?= =?us-
ascii?q?qBAQsBgSkJKIFRCAIrLTsYLRsBjTWffIsLhieBIOZKgwSFMRMIBwEBAQEB.
```

**SENDER-REPUTATION: 3.5**

6. What does the X-Sender-Reputation mean?

**The sender reputation is a calculation based on if the sender has sent spam emails or has a low response rate. It is used to determine the validity of the sender or the messages they send.**

7. What is the name and IP address of the first server to process the email?

```
Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256_P256) id
15.1.2507.17; Sun, 5 Feb 2023 13:30:19 +0100 Received: from SRV-
MAIL.ccrlp.local ([fe80::29bd:50d4:8ee6:5150]) by SRV-MAIL.ccrlp.local
([fe80::29bd:50d4:8ee6:5150%12]) with mapi id 15.01.2507.017; Sun, 5 Feb 2023
13:30:19 +0100 From: COMMUNICATION - CCRLP <communication@ccrlp.fr> To:
```

**Received: from SRV-MAIL.ccrlp.local ([fe80::29bd:50d4:8ee6:5150]) by  
SRV-MAIL.ccrlp.local([fe80::29bd:50d4:8ee6:5150%12]) with map id 15.01.2507.017;  
Sun, 5 Feb 2023 13:30:19 +0100**

8. What is the subject of the email?

```
MAIL.ccrlp.local ([fe80::29bd:50d4:8ee6:5150]) by SRV-MAIL.ccrlp.local
([fe80::29bd:50d4:8ee6:5150%12]) with mapi id 15.01.2507.017; Sun, 5 Feb 2023
13:30:19 +0100 From: COMMUNICATION - CCRLP <communication@ccrlp.fr> To:
"in@ccrlp.fr" <in@ccrlp.fr> Subject: FW: Vital--!!! Thread-Topic: Vital--!!!
Thread-Index: Adk5Wrrd/QrWeukPKE+ATkqmYGz81wAAAYOQAACj1PA= Date: Sun, 5 Feb
2023 12:30:19 +0000 Message-ID: <051e61f3ca4f4893984b0fb809690fa4@ccrlp.fr>
Reply-To: COMMUNICATION - CCRLP <communication@ccrlp.fr> Accept-Language: fr-
```

**Subject: FW: Vital--!!! Thread-Topic: Vital--!!!**

9. What email address does the sender want the receiver to respond to?

human resources department to give out jobs to new students in your university. To Apply, kindly email your Full Name | Age | Address | Alternate Email (different from school email) and mobile number to my email below.

**NOTE:** Your information must be sent to the following email only **drvic341@gmail.com** to ensure swift response.

The email designed to be responded to is: **drvic341@gmail.com**

10. Provide a possible reason for the sender choosing this email address.

**The sender is choosing this secondary email to avoid getting flagged by the spam filter. This method is also to prevent the recipient from reporting the directed email.**

11. Can you find the address of the GMU student who received the email? Why or why not?

**It could be either talaria2.gmu.edu or caduceus2.gmu.edu. If it is neither of them, it is because the antispam email had intercepted the message and redirected the email to a spam vault.**

12. Who is the sender of the email?

```
15.1.2507.17; Sun, 5 Feb 2023 13:30:19 +0100 Received: from SRV-MAIL.ccrlp.local ([fe80::29bd:50d4:8ee6:5150]) by SRV-MAIL.ccrlp.local ([fe80::29bd:50d4:8ee6:5150%12]) with mapi id 15.01.2507.017; Sun, 5 Feb 2023 13:30:19 +0100 From: COMMUNICATION - CCRLP <communication@ccrlp.fr> To: "in@ccrlp.fr" <in@ccrlp.fr> Subject: FW: Vital--!!! Thread-Topic: Vital--!!! Thread-Index: Adk5Wrrd/QrWeukPKE+ATkqmYGz81wAAyOQAACj1PA= Date: Sun, 5 Feb 2023 12:30:19 +0000 Message-ID: <051e61f3ca4f4893984b0fb809690fa4@ccrlp.fr> Reply-To: COMMUNICATION - CCRLP <communication@ccrlp.fr> Accept-Language: fr-FR, en-US Content-Language: en-US X-MS-Has-Attach: yes X-MS-TNEF-Correlator:
```

**From: COMMUNICATION - CCRLP <communication@ccrlp.fr> To:**

**"in@ccrlp.fr" <in@ccrlp.fr>**